

We investigate positive solutions of $-\Delta u + \tau^2 u \equiv g(u)$ for $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(z) \in \mathcal{O}(z^\alpha)$, $\alpha > 0$ non-decreasing.

Theorem. *Let u be a C^2 solution of the above, tending to 0 at infinity. Then u is radially symmetric about x_0 .*

PROOF SKETCH. WLOG $\tau^2 = 1$. $u, |\nabla u| \in \mathcal{O}(e^{-|x|})$, $f(x) := g(u(x)) \in \mathcal{O}(e^{-\alpha|x|})$, $\alpha > 1$. Comparison with $\phi(x) := G(r)(1 + 1/r^s)$, $0 < s$ small yields $|x|^{\frac{n-1}{2}} e^{|x|} u(x) \geq \mu_0 > 0$ when $|x| \gg 0$. Passing to the limit,

$$\sqrt{\frac{\pi}{2}} \int_{\mathbb{R}^n} f(y) e^{\xi \cdot y} dy \geq \mu_0, \quad n = 3 : G(x) = \frac{e^{-\alpha|x|}}{4\pi|x|} \quad (1)$$

STEP 1. $(\exists \Lambda > 0)(\forall x_1 \geq \Lambda \implies u_{x_1}(x) < 0)$. Otherwise, $\exists(x^i)$ with $x_1^i \rightarrow +\infty$ and $u_{x_1}(x^i) \geq 0$. Whence,

$$\frac{|x^j|^{\frac{n+1}{2}} e^{|x^j|} u_{x_1}(x^j)}{x_1^j} \rightarrow -\sqrt{\frac{\pi}{2}} \int_{\mathbb{R}^n} f(y) e^{\xi \cdot y} dy$$

STEP 2. $(\exists \lambda_0 \geq \Lambda)(\forall \lambda \geq \lambda_0)(x_1 < \lambda \implies u(x) < u(x^\lambda))$. Instead, $\exists(\lambda^i) \rightarrow \infty$, $\exists(x^i), x_1^i < \lambda^i$ s.t. $u(x^i) \leq u(x^i)$. In fact, $x_1^i < \Lambda$ and $|x^i| \rightarrow \infty$. If $\xi_1 < 0$, $d := |z^j| - |x^j| \implies d^2 + 2d|x_1^j| = 4\lambda^2 - 4\lambda x_1$ and $x_1 \sim |x^j| \xi_1$ implies $d \rightarrow \infty$. One can show that $|x^j|^{\frac{n-1}{2}} e^{|x^j|} u(x^j) \rightarrow 0$.

If $\xi_1 = 0$, using Step 1, if $\lambda^i \geq \lambda \geq \Lambda$: $u(x^j) \leq u(z^j) \leq u(x^{i^\lambda})$. Then,

$$0 \geq \lim \frac{|x^j|^{\frac{n+1}{2}} e^{|x^j|}}{\lambda - x_1^j} \omega_\lambda(x^i) = \sqrt{2\pi} \int_{\mathbb{R}^n} f(y) e^{\xi' \cdot y'} (\lambda - y_1) dy$$

STEP 3. The collection of λ satisfying the condition in *Step 2* is open. If not, for λ in this set, $\exists(\lambda^i) \rightarrow \lambda$, $(x^i) \rightarrow \infty$, $x_1^i < \lambda^i$ and $u(x^i) \leq u(z^i)$. If $\xi \leq 0$ then noting $|x^\lambda| - |x| \equiv \frac{4(\lambda - x_1)\lambda}{|x^\lambda| + |x|}$ we see $|x^j| - |z^j| \rightarrow 2\lambda\xi_1$ and

$$\int_{\mathbb{R}^n} f(y) e^{\xi \cdot y} dy \leq e^{2\lambda\xi_1} \int_{\mathbb{R}^n} f(y) e^{\xi^0 \cdot y} dy \quad (2)$$

In $x_1 < \lambda$, $w_\lambda := u - u_\lambda > 0$ and using $g \uparrow$: $(\Delta - 1)w = g(u_\lambda) - g(u) \leq 0$. Using the comparison function $\psi := -\sqrt{\pi/2}(e^{-r}/r^{\frac{n+1}{2}})x_1(1 + \mathcal{O}(1/r))$ we see $\exists \mu > 0$

$$\frac{|x|^{\frac{n+1}{2}} e^{|x|}}{\lambda - x_1} \omega_\lambda \geq \mu > 0 \quad (\star)$$

Case $\xi_1 = 0$; passing to the limit of the left hand in $(\star)$ yields $\int_{\mathbb{R}^n} e^{\xi \cdot y} f(y)(\lambda - y_1) dy \leq 0$ which contradicts $(\star)$. Where $\xi_1 < 0$, $\frac{|x^j|}{\lambda - x_1^j} < \frac{-|x^j|}{x_1^j}$ implies

$$\frac{-1}{\xi_1} \int_{\mathbb{R}^n} (e^{\xi \cdot y} - e^{2\lambda\xi_1 + \xi^0 \cdot y}) dy \geq \mu$$

STEP 4. Suppose $\exists \lambda$ s.t. $x_1 < \lambda$ implies $u \geq u_\lambda$ and $u \not\equiv u_\lambda$. Then $u > u_\lambda$ whenever $x_1 < \lambda$. To see this, apply the *Maximum Principle* to $z := u - u_\lambda \geq 0$ and $L[\cdot] := (-\Delta - 1)(\cdot) \geq 0$.

There exists a maximal interval (γ_1, γ_2) where we have this property for $\lambda \in (\gamma_1, \gamma_2)$. Obviously, $\gamma_1 \neq -\infty$ and we must have $u \equiv u_{\gamma_1}$ in Σ_{γ_1} since otherwise we contradict our choice of maximal interval by continuity. $\square$